MIL-C-25601C (USAF)
16 October 1969
Superseding
MIL-C-25601B (ASG)
10 April 1962

MILITARY SPECIFICATION

COUPLER, ANTENNA
CU-308D/U

1. SCOPE

1.1 This specification covers a multi-band longwire antenna coupler.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS:

MILITARY

MIL-C-17/28	Cable, Radio Frequency Coaxial
MIL-C-172	Cases; Bases, Mounting, and Mounts, Vibration (for use with Electronic Equipment in Aircraft)
MIL-C-3608	Connectors, Coaxial, Radio Frequency, Series BNC, and Associated Fittings, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-R-7276	Receiving Set, Radio AN/APN-70B
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-C-39012	Connectors, Coaxial, Radio Frequency, General Specification for

STANDARDS:

MILITARY

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment

FSC 5985

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MIL-STD-471 Maintainability Demonstration

MIL-STD-781 Reliability Tests Exponential Distribution

MIL-STD-831 Test Reports, Preparation of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

3. REQUIREMENTS

Preproduction - This specification makes provision for preproduction testing.

3.2 General specification - The requirements, including material and workmanship, specified in Specification MIL-E-5400 for Class 2 equipment are applicable as requirements for this specification with the additions or exceptions to the paragraphs of Specification MIL-E-5400 as identified by the similarity of paragraph headings. If any of the requirements of this specification conflict with the requirements of Specification MIL-E-5400, the requirements of this specification shall govern.

3.2.1 Separable component parts - Component parts such as subassemblies shall be located in the equipment and attached thereto so that they are readily accessible and can be replaced without the use of any special tools. All separable component parts shall be so designed that they can be inserted in the equipment in but one position, that position being the one for optimum operation and performance.

3.2.2 Circuits - All circuits shall be so designed that no damage occurs when the equipment is operated with any combination of operating controls and service adjustments set to any possible combination of settings.

3.2.3 Fuse - Fuses shall not be included in the equipment. Fusing shall be provided in the aircraft wiring system.

3.2.4 Convenience - The design of the major subassemblies shall be such that they may be removed from the chassis proper, without unsoldering any connections. After being removed, it shall be possible to interconnect them with the chassis to form a complete working umit for test purposes.

3.3 Design and Construction.

3.3.1 General - No relays, or other electromechanical devices shall be used. The design shall be completely "solid state". No electronic tubes shall be used. Where possible the semiconductor devices used, shall be selected from established military preferred types.

3.3.1.1 Assumptions - Design and utilization of the coupler will be based on the following assumptions:

a. That the antenna transmit receiver relay will be provided

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in other equipment and that said other equipment will not present more than 100K loading on the antenna at the LORAN frequencies.

b. That the LORAN equipment to be used in conjunction with this coupler will provide two control lines. One of the two control lines will select the LORAN "A" or the LORAN "C" function by the grounding of that line or the ungrounding of that line, respectively. The other control line will select the local or distance gain characteristic of the amplifiers by the grounding or ungrounding of that control line respectively.

3.3.2 Maintenance Provisions

3.3.2.1 Adjustment provisions - Provision shall be made for adjusting the antenna capacitance to a constant value regardless of variations in the various aircraft installations provided that the fixed installation capacitance falls within the specified limits. This adjustment shall be labeled "ANT". A screw cap using a conducting gasket shall be provided on the front panel for access to this adjustment. The adjustment itself shall be sturdy enough to withstand repeated use and require only a standard slot type screwdriver. The screw itself shall be at ground potential. No maintenance adjustments shall be required under flight conditions.

3.3.2.2 Service bench operation - When the coupler CU 308D/U is removed from its housing, it shall be possible to place it on any side except front and rear on a smooth surface without causing damage to any components. It shall be possible to operate it when it is removed from its housing, but the detailed performance specified herein need not be provided while the housing is removed. It shall also be possible to service and align the coupler when removed from the main unit without objectionable interference upon any r-f channel.

3.3.3 Diodes - Diodes shall be selected in accordance with MIL-S-19500.

3.3.4 Sealing - The coupler unit shall be built in a metal container with a conducting gasket seal, which will prevent breathing at altitudes from sea level to 70,000 feet when sealed. Provisions shall be made for opening and resealing the case for maintenance purposes. The air enclosed in the sealed unit shall be free of gas vapors which would cause deterioration of the enclosed components of the equipment.

3.3.5 Radio frequency connectors - BNC series type UG-657/U or equivalent shall be used for RF connections. Connectors shall be chosen in accordance with MIL-C-39012 or MIL-C-3608.

3.3.6 Power connector - A satisfactory locking device shall be provided on the power connector to prevent accidental disengagement of the plug.

3.3.7 Mounting - The couplers shall be constructed for use with the MT-1094/U mounting device. This mounting, Fig 7 shall provide for installation

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in any altitude. The mounting is in accordance with the general requirements of specification MIL-C-172 and shall permit operation of the coupler under the shock and vibration conditions of Specification MIL-E-5400. The coupler shall be designed to facilitate installation and removal from its mounting. No special tools shall be required for this purpose.

3.3.8 Size - The dimensions of the coupler, Fig 7, shall conform to the requirements of 3.3.12(f) and 3.3.7. Mechanical interchangeability shall include identical placement of mounting screws, connectors and access plug as well as full compatibility with the MT-1094/U shock mount.

3.3.9 Weight - The weight of the coupler shall be kept to a minimum and shall not exceed two pounds ten ounces including the shock mount.

3.3.10 LORAN coax ground loop isolation - The LORAN amplifier output jack on the coupler shall be mounted on an insulated washer and the secondary windings of the LORAN amplifiers output transformers shall be connected to it with no connection to the coupler chassis ground. The purpose of this requirement is to prevent power conduction through the coax shield between two different ground points in the vehicle in which the equipment is installed.

3.3.11 Power source - The design of the coupler shall be such that all operating power shall be obtained from a 27.5 volts DC aircraft battery supply. No degradation in coupler performance shall be noted between the operating power input voltages of 21 and 33 volts DC. No other power source is required.

3.3.12 Functions - The coupler shall provide the following functions:

a. Filter out undesirable frequencies above and below the LORAN bands.

b. Amplify all communications frequencies from two to thirty MHz and provide a 50 OHM output capable of driving two "Comm" receivers by means of two coaxial "Comm" jacks, and up to 120 feet of RG-58/U cable.

c. Suppress high intensity static and electrostatic discharges by voltage limiting devices.

d. Provide simultaneous operation of the LORAN receiver and one or two Comm receivers.

e. Amplify the LORAN "A" and "C" frequencies and supply both to a common 50 OHM output jack with sufficient output power to drive the APN-70B receiver input (see MIL-R-7276) through a 120 ft length of RG-58/U cable.

f. Directly replace, mechanically and electrically all equipment previously covered by this specification.

NOTE: No aircraft wiring changes shall be required when this equipment is

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plugged in to replace older units, or to add the LORAN "C" function to aircraft equipped for "A" only, using the APN-70A equipment.

g. This coupler shall provide means for simultaneous matching of the antenna to the LORAN "A", LORAN "C" and "COMM" amplifiers contained within the coupler unit. The antenna matching circuitry shall provide an optimum balance between sensitivity, bandpass, and signal to noise ratio.

3.4 Government loaned property - When provided for in the contract or purchase order, one complete CU308D/U will be furnished by the government on loan to the contractor upon his request.

3.5 Electrical Performance

3.5.1 Bandwidth - The bandpass properties of the coupler shall be determined with the equipment connected as shown in Figure 1. The bandpass characteristics of the LORAN "A" and the LORAN "C" function in both the local and distance modes shall be as specified in Table 1. The bandpass characteristics of the Comm amplifier shall be as specified in 3.5.4. All measurements shall be made under the standard test conditions specified in 4.3.

3.5.2 LORAN signal reception sensitivity - The sensitivity of the LORAN "A" mode of operation is such that not greater than five microvolts at the antenna terminal of 1000 cycle 100 percent modulated carrier shall produce a five to one signal plus noise to noise ratio at the receiver indicator screen. In the LORAN "C" mode of operation no greater than ten microvolts are required for the same results. Connect the equipment as in Figure 4.

3.5.3 Communications amplifier reception sensitivity-The communications amplifier sensitivity shall be such that a two microvolt input signal within the frequency range of two to thirty megahertz shall produce a minimum of 6 db signal plus noise to noise ratio, as measured at either Comm output terminal of the coupler. Arrange the equipment as shown in Figure 2.

3.5.4 Communications amplifier signal loss/gain - The communications amplifier section of the coupler shall exhibit gains and losses in accordance with the voltage ratios as given in Table No. II. At the frequencies where the insertion loss limits change, the greater of the two loss figures shall apply. Connect the equipment as shown in Figure 2.

3.5.5 Antenna input impedance - With the equipment connected as shown in Figure 1 the LORAN "C" frequency input impedance shall be such that when the test switch is opened and closed across the 120K resistor no greater than 20 db change in output from the coupler shall occur. Input impedances for the LORAN "A" frequency band and the communications receiver band are given in Table III. Connect the equipment as shown in Figure 3.

3.5.6 Power consumption - The power drain shall not exceed 125 ma between 21 and 33 volts at the coupler power connector.

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3.5.7 Antenna capacitance tuning ranges - The coupler shall be capable of tuning all antenna capacitances between 30 and 275 pf.

3.5.8 Channels - Each coupler shall be capable of LORAN reception at the following nominal frequencies:

Channel 1	1950KHz	Channel 4	1750KHz
Channel 2	1850KHz	Channel 7	100KHz
Channel 3	1900KHz		

3.5.9 Intermodulation Spurious Response

3.5.9.1 Communications amplifier - Two unmodulated 50MV signals at frequencies of two and seven MHz shall produce a spurious response no greater than an equivalent 100 microvolt signal at the difference frequency of five MHz per second. The equipment shall be connected as in Figure 5.

3.5.9.2 LORAN amplifier - HF channels - Two unmodulated 100 MV signals at frequencies of three MHz and 1.1 MHz shall produce a spurious response no greater than an equivalent ten UV signal at the difference frequency of 1900 KHz (80 db). Connect the equipment as shown in Figure 6.

3.5.9.3 LORAN amplifier (LF, 100 KHz channel) - Two unmodulated 200MV signals at frequencies of 200KHz and 300KHz at the coupler antenna terminal shall produce a spurious response no greater than an equivalent ten UV signal at the difference frequency of 100KHz (86 db). Connect the equipment as shown in Figure 6.

3.5.10 Dynamic Input Range

3.5.10.1 Communications amplifier - The dynamic input range of the communications amplifier shall be such that intelligible voice communications may be received throughout the 114 db range between two UV and one V. This may be checked with equipment arranged according to Figure 2 at a comm frequency of seven MHz using 100 percent modulated 1000 cycle carrier, and head phone monitoring.

3.5.10.2 Each LORAN amplifier system shall be capable of supplying visually undistorted 1000 cycle sinusoidal 100 percent modulated signals to the LORAN receiver between their respective sensitivity ratings given in 3.5.2 and a signal strength 106db higher. Set up equipment as in Figure 4.

3.5.11 Off selection sensitivity on LORAN amplifier - When the LORAN A amplifier is selected at the LORAN receiver, the LORAN C amplifier in the coupler shall exhibit 80 db minimum less gain than it does when operated in the distance mode and is selected at the LORAN receiver. When the LORAN C amplifier is selected at the LORAN receiver, the LORAN A amplifier in the coupler shall exhibit 80 db minimum less gain than it does when selected at the LORAN receiver, and operated in the distance mode.

3.5.12 Local distance mode attenuation factors - In the LORAN A and LORAN C Mode, the local distance switch shall cause a minimum of 30 db attenuation of the signal when transferred from the distance to the local

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position and shall cause a maximum of 40 db attenuation under this same condition.

3.5.13 LORAN coax ground loop isolation - The maximum coupling between the LORAN jack shield connection and the coupler chassis shall be 150 pf at 100 KHz and one meg ohm DC.

3.5.14 Interference. Interference control requirements shall be in accordance with MIL-STD-461.

3.5.15 Antenna terminal protection - The following currents may flow into the antenna terminal without damage to the coupler:

Plus or minus 5 amps for 10 seconds.
Plus or minus 25 amps for 1 second.
Plus or minus 50 amps for .1 second.

A one minute cooling period shall be allowed between tests.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the government. Inspection records of the examination and tests shall be kept complete, and available to the government as specified in the contract or order. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests - The inspection and testing of couplers shall be classified as follows:

a. Preproduction tests

b. Quality conformance inspection tests

4.3 Standard test conditions - Unless otherwise specified, the couplers shall be tested under the following conditions:

a. Ambient temperature - Room ambient, between the limits of plus 15 degrees and plus 35 degrees C (plus 59 degrees and plus 95 degrees F).

b. Humidity - Prevailing ambient (between 10 and 95 percent).

c. Atmospheric pressure: Normal

d. Interconnecting cables: Interconnecting cables between the coupler and the communications receivers or LORAN receivers shall be 100 feet.

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e. Primary power supply - All detail testing shall be accomplished with the line voltages maintained at 27.5 DC, measured across the input terminals at the coupler end of the primary power supply cable.

f. Transistors - The transistors used shall be selected at random from representative groups of the selected types.

g. Warm-up period - A maximum of five seconds for each test.

h. Orientation - Substantially upright.

i. Tuning - The antenna tuning capacitor shall be set for a 160 pf antenna. This shall be referred to as the nominal test capacitence.

j. The COH DET switch on the R277C/APN70B shall be "ON" (down).

k. All coaxial lines shall be RG-58/U per MIL-C-17/28.

4.3.1 Test conditions (Comm amplifier) - Unless otherwise specified, the following test conditions in addition to the standard test conditions specified in 4.3 shall apply during all tests and performance measurements of the communications amplifier portion of the coupler.

a. Dummy antenna - The output of the standard signal generator shall be connected to the input of the coupler in parallel with the nominal test capacitance specified in 4.3(i), initial input voltage calibrations shall be made at the input terminal of the coupler.

b. Standard input signal - To be obtained from a standard signal generator, Measurements Co Model 65B or equal.

c. Radio test receiver - Empire devices NF 105 or equal shall be calibrated and operated as directed in the instruction manual for that equipment.

4.3.2 Test conditions - LORAN - Unless otherwise specified, the following test conditions in addition to the standard test conditions specified in 4.3 shall apply during all tests and performance measurements of the LORAN portion of the coupler.

4.3.2.1 Coupler output load - The coupler output load on the coupler through the transmission line shall consist of a 50 ohm resistor at the end of 100 feet of RG-58/U coaxial cable.

4.3.2.2 Standard signal source - The standard signal source shall be obtained from Measurements Co. Model 65B or equal. The signal shall be unmodulated continuous wave unless otherwise specified.

4.3.2.3 Receiver gain control - The receiver gain control shall be set up as required.

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4.3.3 Test conditions (intermodulation - spurious response) - unless otherwise specified, the test conditions for the comm amplifier and LORAN reception outlined in 3.5.9 and in Figures 5 and 6 in addition to the standard test conditions in 4.3 shall apply during all intermodulation (spurious response) measurements of the coupler.

4.4 Preproduction testing

4.4.1 Preproduction test samples - The preproduction tests samples shall consist of models representative of the production equipment. They shall be tested under the test conditions specified herein and at the point designated by the contract.

4.4.1.1 Sampling instructions - The contractor shall conduct tests on eleven preproduction sample equipments (or less, dependent on contractual requirements) to determine that the design of the equipments meet the requirements specified herein. The contractor's test shall be conducted in accordance with the test procedures specified herein.

4.4.1.1.1 Testing schedule - The eleven sample equipments shall be processed as follows:

a. Sample No. 1 - Sample No. 1 shall be subjected to the following tests:

Tests	Applicable Paragraph
Examination of product	4.6.1
Operational tests	4.6.2
Environmental tests as follows:	4.6.3
Shock	4.6.3.5

b. Sample No. 2 - Sample No. 2 shall be subjected to the following tests:

Tests	Applicable Paragraph
Examination of product	4.6.1
Electrical tests	4.6.2.1
Environmental tests as follows:	4.6.3
Temperature-altitude	4.6.3.3
Humidity	4.6.3.6
Vibration	4.6.3.4
Equipment stability	4.6.4

c. Sample No. 3 - Sample No. 3 shall be subjected to the following tests:

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Tests	Applicable Paragraph
Examination of product	4.6.1
Electrical tests	4.6.2.1

d. Seven equipments shall be subjected to the reliability demonstration test specified in 4.6.5. Two of the seven equipments shall be continually tested for longevity as specified in 4.6.6.

e. One equipment shall be subjected to the maintainability demonstration test specified in 4.6.7. The equipment shall be subjected to examination of product, 4.6.1, and electrical tests, 4.6.2.1, prior to, and after, the maintainability demonstration test.

4.4.2 Preproduction tests - Preproduction tests shall consist of the tests specified under 4.6, subjected to the test conditions specified herein.

4.5 Quality conformance inspection tests - Quality conformance inspection tests shall consist of individual test and sampling plans and tests.

4.5.1 Individual tests - Each coupler, CU-308D/U shall be subjected to the following tests:

Tests	Applicable Paragraph
Examination of product	4.6.1
Electrical tests	4.6.2.1

4.5.2 Sampling Plans and Tests

4.5.2.1 Sampling plan A - Production reliability acceptance tests shall be conducted in accordance with the requirements of test level E of MIL-STD-781. The test program shall be as follows:

Production Reliability
Test Program

Quantity ordered per year per contractor	Number of Reliability test programs per increment *	Total test programs per no. ordered per year
1 to 100	1	1
101 to 200	1	2
Over 200	One from each additional 100 or franction thereof on order	Variable

* A reliability test program shall consist of a quantity of at least four and not more than seven units to be tested. The actual number shall be that which is most economical.

4.5.2.1.1 Accept-reject criteria - Accept-reject criteria shall be in

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accordance with test plan I of MIL-STD-781.

4.5.2.2 Sampling plan B - Quantities of coupler, Antenna CU-308D/U shall be selected at random as outlined below and subjected to the sampling tests of 4.5.2.2.1.

Sampling Plan B

Quantity of Consecutively Produced items on order by Groups Per Contractor	Number tested per increment	Total tested per number ordered per year
1 to 25	1	1
26 to 50	1	2
51 to 75	1	3
76 to 125	1	4
126 to 200	1	5
201 to 300	1	6
Over 300	One from each additional 100 or fraction thereof on order	Variable

4.5.2.2.1 Sampling tests - Sampling tests shall consist of the following:

Tests	Applicable Paragraph
Environmental tests as follows:	4.6.3
Temperature-altitude	4.6.3.3
Vibration	4.6.3.4
Humidity	4.6.3.6

4.5.2.2.2 Rejection and retest - When one equipment from a production run fails to meet the specification, no equipments still on hand or produced later shall be accepted until the extent and cause of failure are determined and corrected.

4.5.2.2.3 Continuation of tests - For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of equipments on hand or produced later shall not be made until it is determined that equipments meet all the requirements of the specification.

4.5.3 Defects in equipments already accepted - The investigation of a test failure could indicate that defects may exist in equipments already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

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4.6.1 Examination of product - The Coupler, Antenna, CU-308D/U shall be visually inspected to verify that the materials, design and construction, necessary mechanical measurements, marking, and workmanship comply with the requirements of this specification.

4.6.2 Operational tests

4.6.2.1 Electrical tests - The procedures and application of electrical tests to determine proper performance and compliance with the requirements of this specification shall be prepared by the contractor and subject to approval by the procuring activity. The electrical tests shall be performed prior to and after the environmental tests.

4.6.2.2 Interference - Compliance with electrical interference and susceptibility requirements shall be checked in accordance with MIL-STD-461.

4.6.3 Environmental tests - Environmental testing shall be in accordance with class 2 requirements of MIL-T-5422 with the following additions or exceptions. When the two documents conflict, this specification shall govern.

4.6.3.1 Exempted tests - Salt spray, sand and dust, and fungus resistance tests shall not apply unless specifically designated for critical units. The explosion test shall not be required on pressure-sealed units.

4.6.3.2 Supplemental procedures - The contractor's test procedures, when approved, shall form a part of this specification.

4.6.3.3 Temperature-altitude test - The coupler shall be subjected to temperature-altitude tests, of Specification MIL-T-5422 with the temperature and altitudes compatible with class 2 requirements of Specification MIL-E-5400.

4.6.3.4 Vibration - The couplers shall be subjected to the vibration tests of Specification MIL-T-5422, Part I.

4.6.3.5 Shock - The couplers shall be subjected to shock tests of Specification MIL-T-5422.

4.6.3.6 Humidity - The couplers shall be tested in accordance with the humidity tests of specification MIL-T-5422.

4.6.3.6.1 Test installation - The equipment shall be placed in a vented test chamber in a manner simulating installed conditions whenever possible. The equipment shall be placed on an inert, interposing material such as loosely woven glass cloth to prevent the development of corrosion due to contact of dissimilar metals. The equipment shall be protected against dripping of condensed moisture by an antidrip device that will not impair the flow of air through the test compartment. An example of this antidrip device is, though not limited to, a gabled or sloped roof.

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4.6.4 Equipment Stability - The coupler shall be subjected to 168 hour test to determine conformance to this specification. The following procedures shall be used:

a. The coupler shall be tuned and loaded in accordance with the requirements specified in 4.3.

b. The cycle duty shall be in accordance with the following procedure: (Alternately in the "C" and "A" mode).

(1) Twenty-five (25) seconds of operation maximum sensitivity (remote operation).

(2) Twenty-five (25) seconds of operation minimum sensitivity (local operation).

(3) Ten (10) seconds power off.

c. The equipment shall be tested for conformance with 3.5.

4.6.5 Reliability demonstration test - The reliability demonstration test shall be conducted in accordance with test level F of MIL-STD-781. Accept-reject criteria plan III for quality conformance inspection tests.

4.6.6 Longevity test - An accumulable longevity test shall be conducted in accordance with the requirements of 4.6.5.

4.6.7 Maintainability - Maintainability tests and demonstration shall be in accordance with MIL-STD-471. The quantitative requirements shall be demonstrated to a confidence level of 90 percent.

4.7 Design data and material

4.7.1 Preproduction test report - The contractor shall prepare a preproduction test report in accordance with MIL-STD-831.

5. PREPARATION FOR DELIVERY

5.1 The Coupler, Antenna CU-308D/U shall be prepared for delivery in accordance with the instructions of the procuring activity.

5.2 Reinspection markings - Reinspection markings shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use - The Coupler, Antenna CU-308D/U covered by this specification is intended for use in aircraft to match impedances, simplify signals, reject near band interference, etc., with LORAN receiving and high frequency communication systems.

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6.2 Ordering data - Procurement documents should specifiy the following:

a. Title, number, and date of this specification.

b. Stock number.

c. Preproduction samples and test.

(1) Number of preproduction samples.

(2) Point of inspection.

(3) Requirement for concurrent delivery of each sample and its test data. The procuring activity should be given at least ten days prior notice when the preproduction tests are to be conducted so that they may be supervised or witnessed, if desired, by a government representative.

d. Selection of applicable levels of preservation and packaging and packing.

Custodians:
Air Force - 85

Reviewer Activity:
Air Force - 11

Preparing Activity:
Air Force - 85
(Project 5985-F305)

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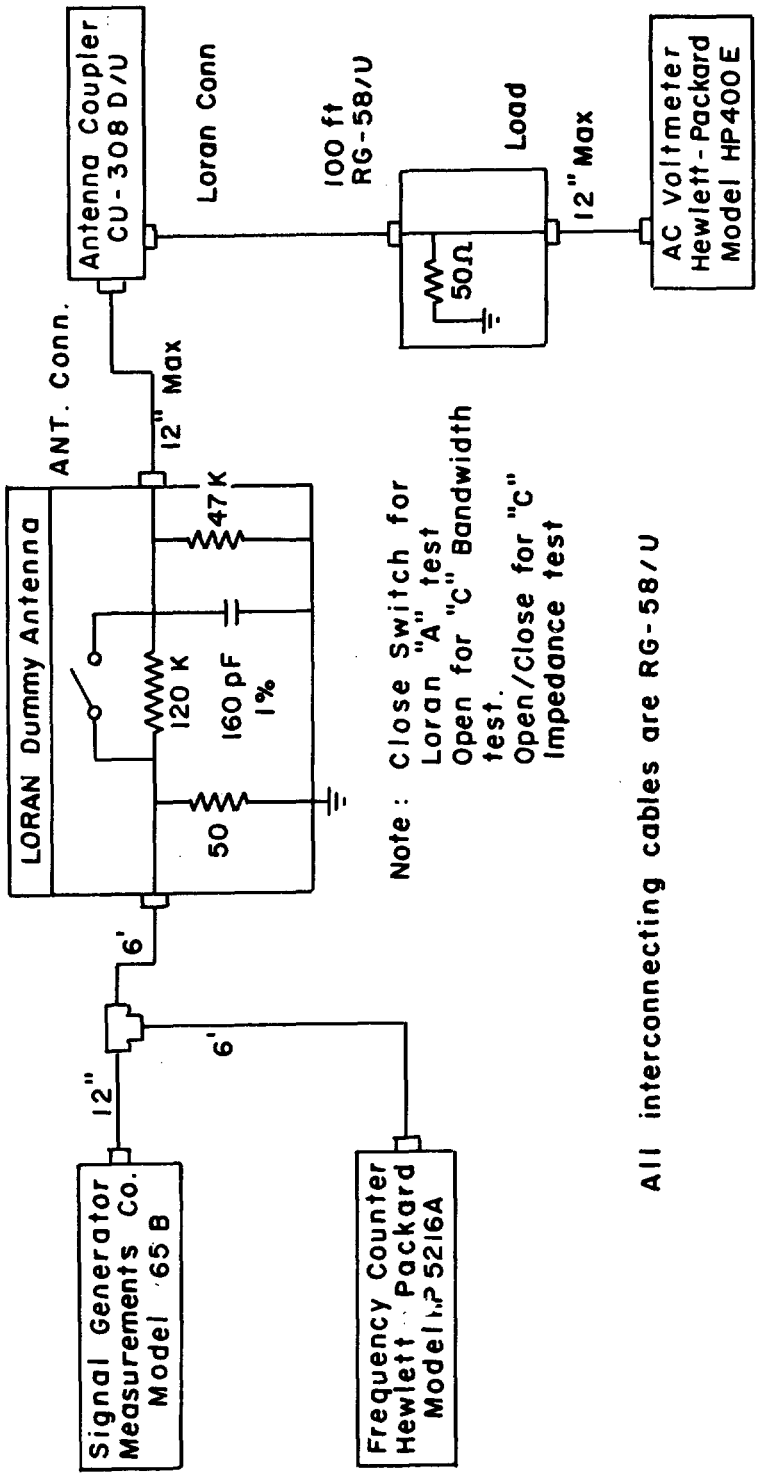


FIGURE 1 LORAN bandpass tests / LORAN "C" Input Impedence

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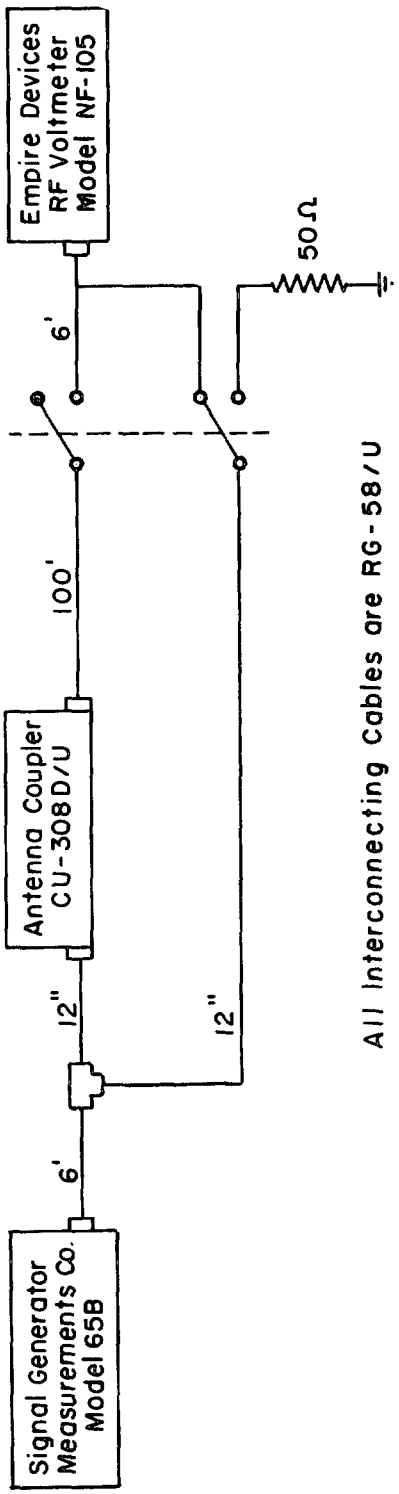


FIGURE 2 Communications Amplifier Bandwidth/Sensitivity Tests

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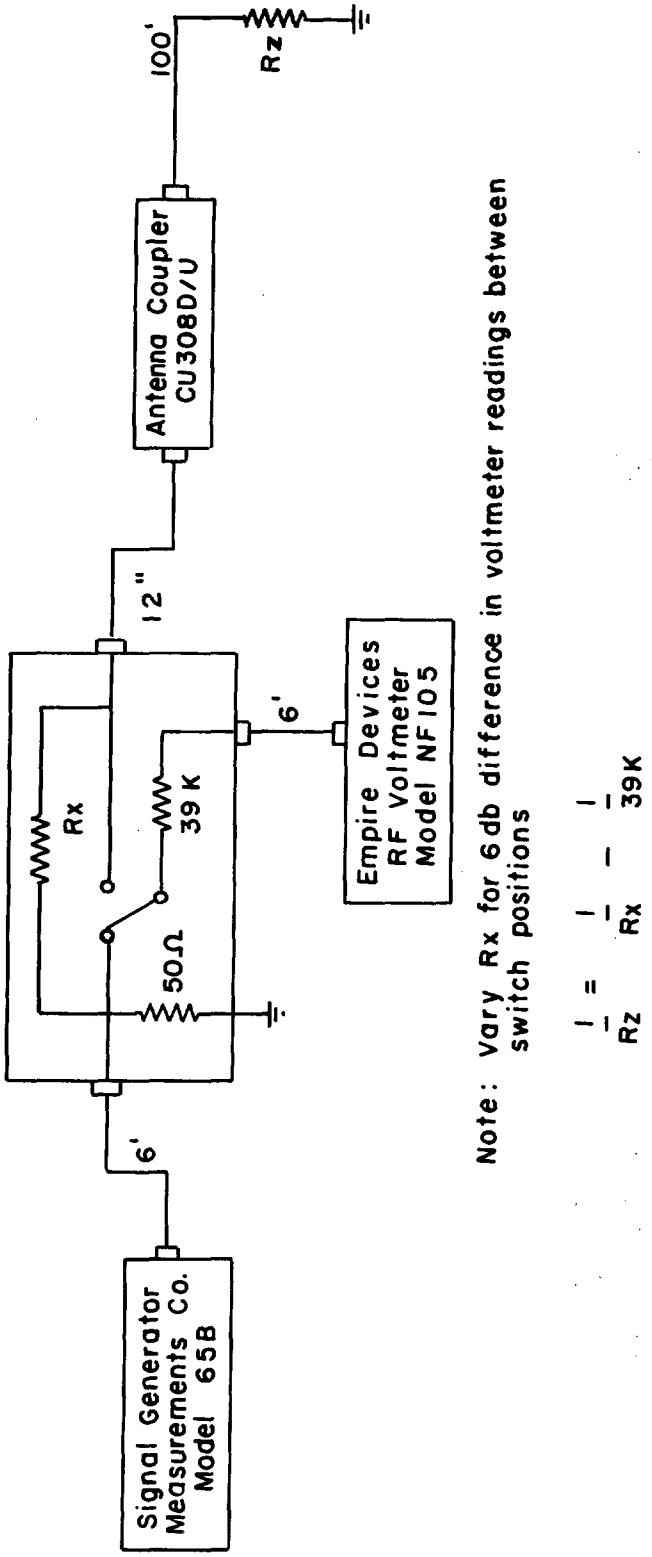
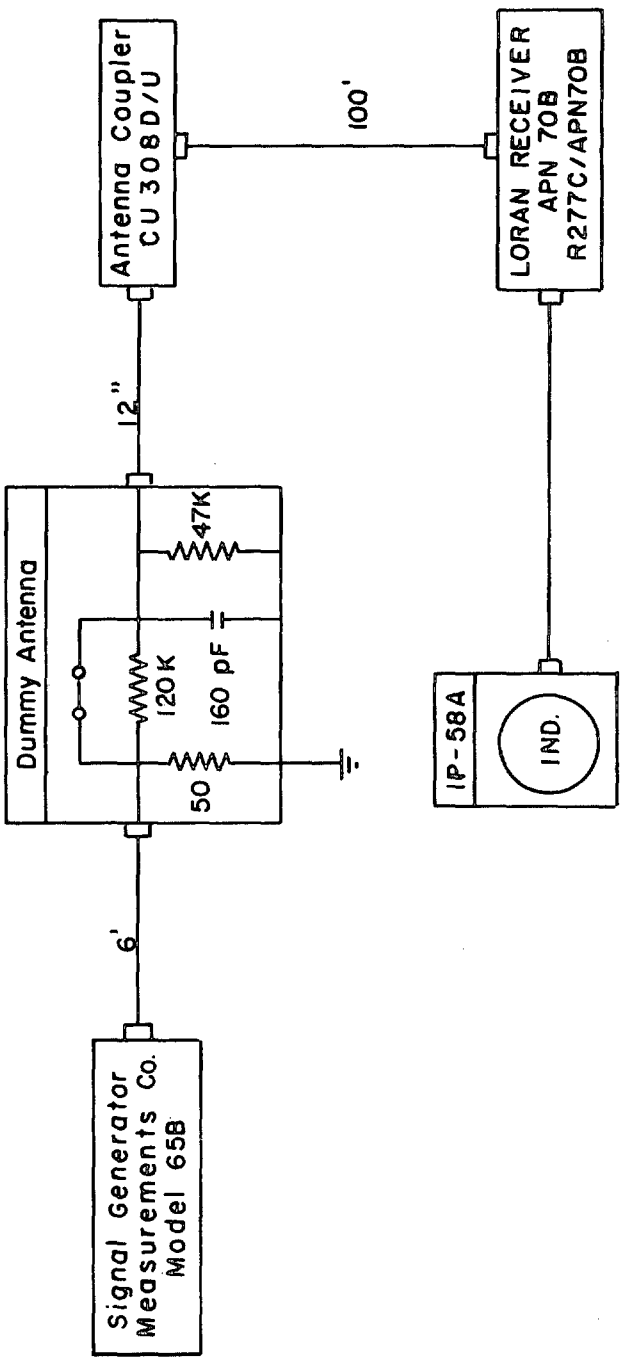


FIGURE 3 Input inpedence tests for Communications Amplifier and Loran "A" frequencies

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NOTES: COH DET switch down and function switch set to 3
Dummy antenna switch closed.
All coaxial lines - RG-58/U

FIGURE 4 Loran sensitivity tests

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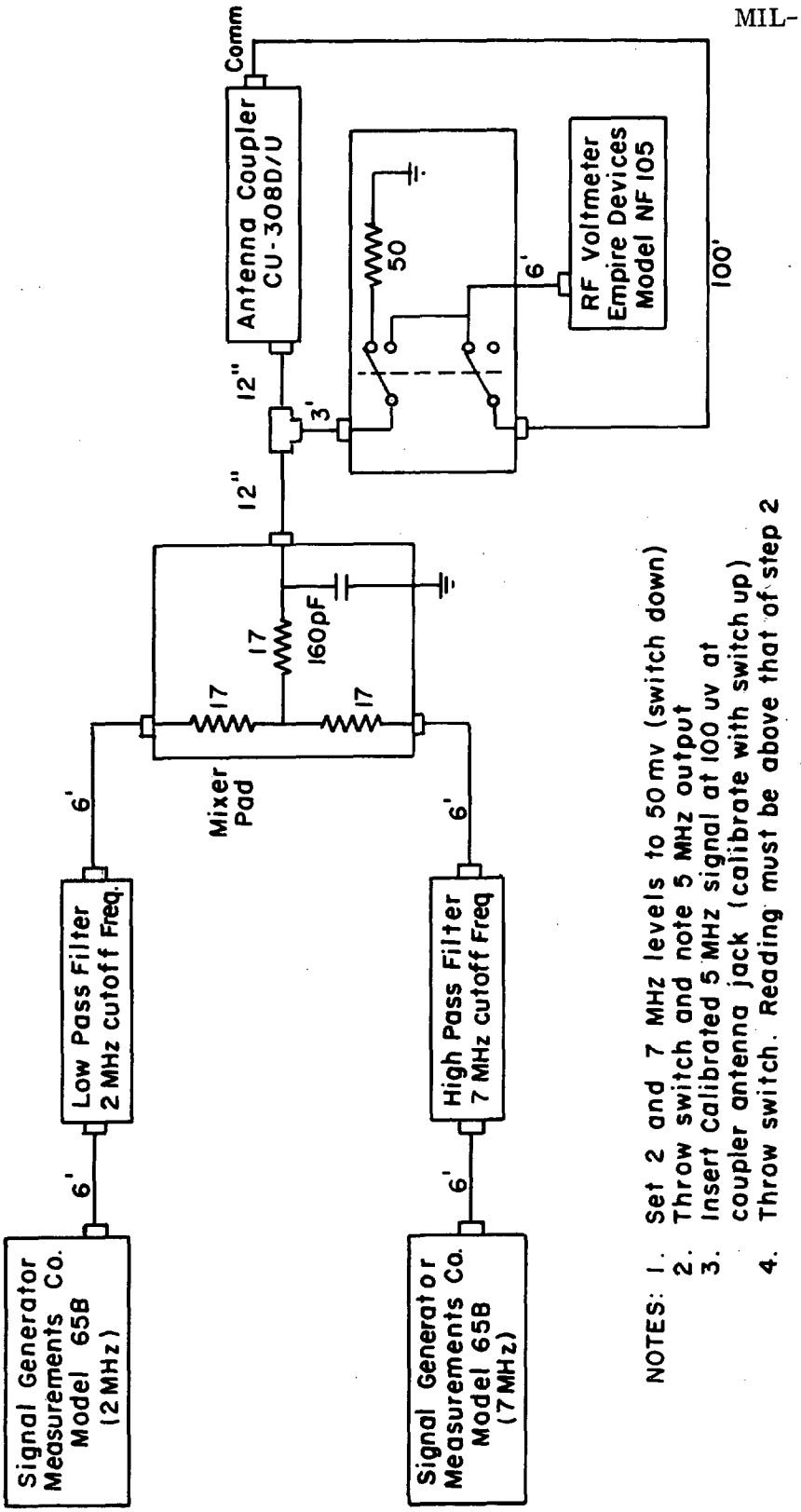


FIGURE 5 Intermodulation Spurious Response Communications Amplifier

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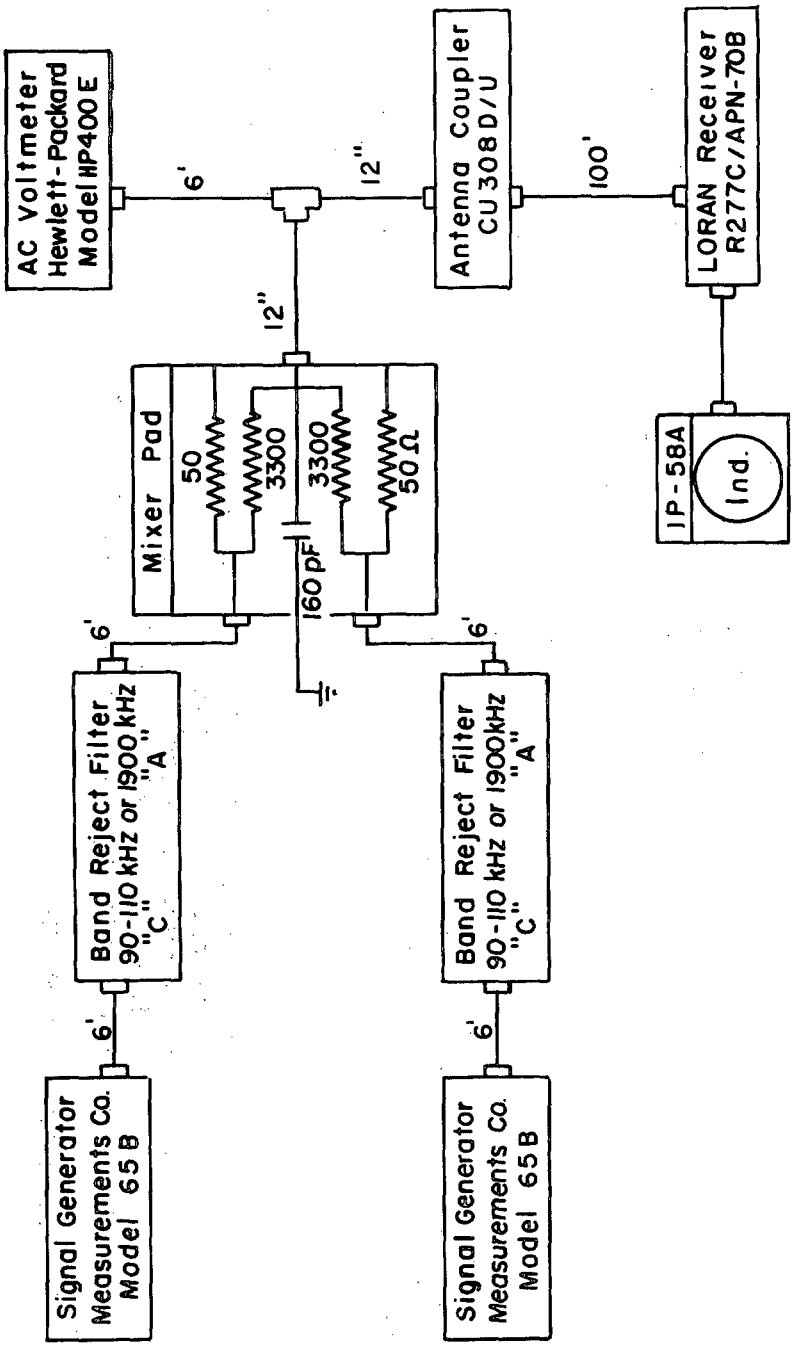
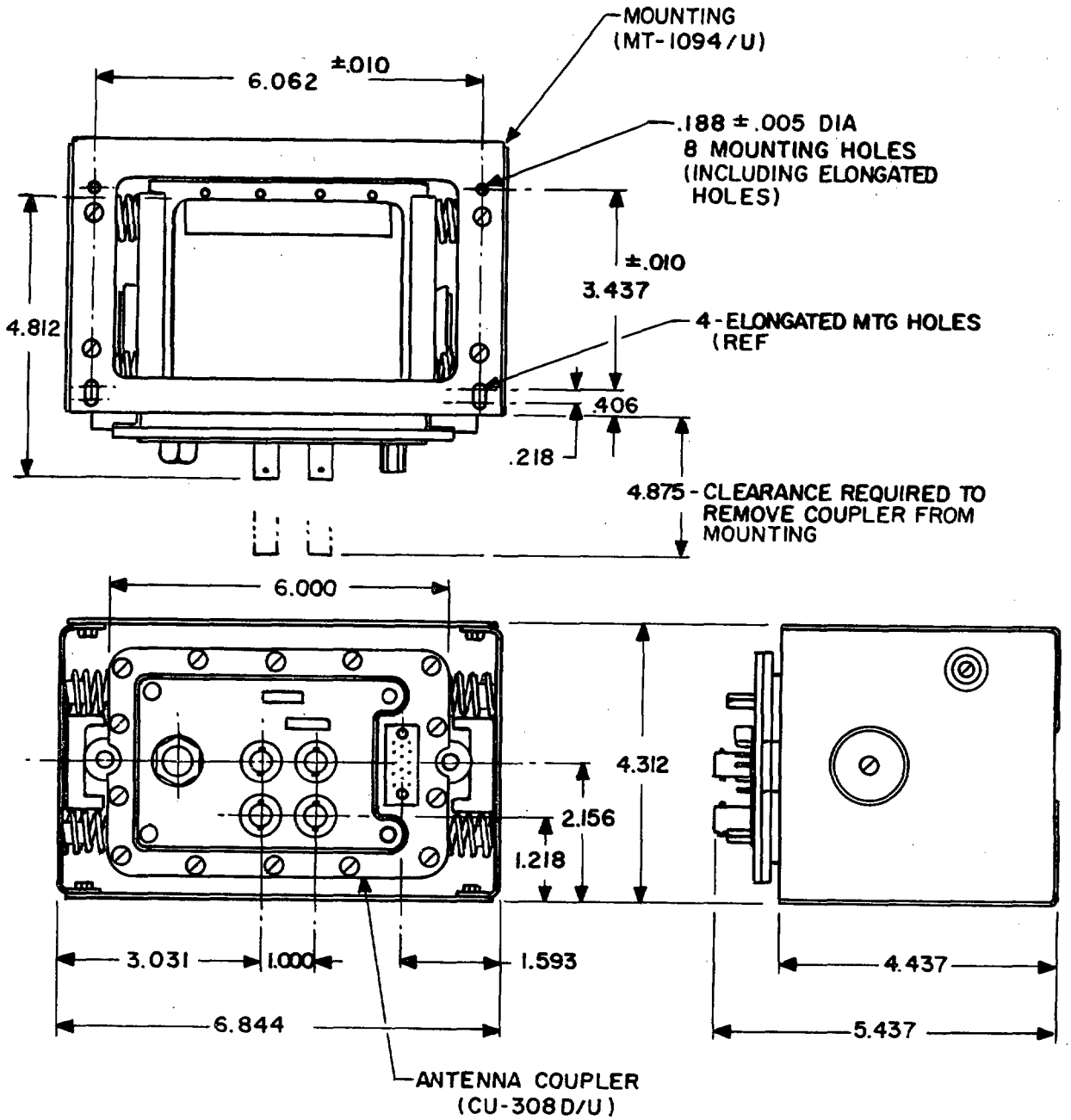


FIGURE 6 Intermodulation Spurious Response for Loran "A" and "C"

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NOTE:
1) DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCES ±.031
2) ALLOW CLEARANCE FOR REMOVAL OF CONNECTORS AND ASSOCIATED WIRING

FIGURE 7 Antenna coupler outline and mounting dimensions

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Table I
Loran Bandpass Characteristics

Channel No.	Mode	Reference Frequency	Minimum +1,-6db Bandwidth	Maximum 40db Bandwidth	Maximum 50db Bandwidth	Maximum 70db Bandwidth	Maximum 90db Bandwidth
7	Local	99.5KHz	92-108KHz	88-112.5KHz	87-114KHz	84-117KHz	65-135KHz
7	Distant	99.5KHz	96-104KHz	89-111KHz	88-112.5KHz	86-115KHz	75-125KHz
1,2,3,4	Local or Distant	1846KHz	1725KHz 1975KHz	1250KHz 2650KHz	1150KHz 3000KHz	N/A	N/A

Table II
Communications Amplifier Bandpass

Frequency	2.7---3.5MHz	3.9---5MHz	5---30MHz	30---40MHz
Minimum output -versus- Input voltage ratio	1/4	1/2	2/1	1/2

Table III
LORAN "A" and Communications Band Input Impedances

Nominal Frequency	2.1MHz	2.4MHz	4.5MHz	10.1MHz	14	21	25	30	1.85("A")
Nominal Impedance ± 20% (in ohm's)	560	470	150	120	82	39	22	15	680

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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